SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Introduction:

An Intelligent agent is a system that observes its environment through sensors and takes actions using actuators to achieve a particular objective. In a smart home automation system, sensors collect information such as motion, temperature, light levels, and time. The system uses this information to control lighting, temperature, and security automatically.

There are four main types of intelligent agents: Simple reflex agents, model-based agents, goal-based agents, and utility-based agents.

1. Simple Reflex Agent

A Simplex reflex agent select actions based only on the current input. It follows condition-action rules such as "if this happens, do that." It does not remember past event.

working on a smart home:

* If motion is detected, the lights turn ON.
* If the room temperature is high, the fan turns ON.
* If an unauthorized person is detected, an alarm is activated.

Advantages: It is simple, fast, and easy to implement.

Disadvantages: It cannot remember previous conditions or learn user preferences. It may make poor decisions when the environment changes.

Example: A motion sensor automically switches on a light when someone enters a room.

2. Model-Based Agent:

A model-based agent maintains an internal model of the environment. It uses current sensor inputs and information about previous two states to understant what is happening.

working in a smart home:

1. It remember whether a room is occupied,
2. It uses temperature readings from previous moments to understant whether the room is becoming warmer.
3. It avoids switching off the lights immediately when a person is temporarily not detected.

Advantages: It can work in partially observable environments and make better decisions using stored information:

Disadvantages: It requires memory and more complex programming. An incorect internal model may lead to incorrect decisions.

Most effective Agent:

A hybrid approach combining model-based, goal-based, and utility-based agents would be the most effective for smart home automation.

The model-based agent maintains information about room occupancy and environmental conditions. The goal-based agent works toward objectives such as maintaining a comfortable temperature and securing the house. The utility-based agent balance comfort, electricity consumption, and security to select the most beneficial action.

A simple reflex, mechanism can also be used for immediate responses, such as activating an alarm when a security sensor detects a threat.

Conclusion:

Different intelligent agents have different capabilities. simple reflex agents are suitable for basic automatic actions, model-based agents use stored information, and goal-based agents plan actions to achieve objectives. utility-based agents balance multiple requirements. Therefore, a hybrid intelligent agent system is the best choice for proriding comfort, energy efficiency, and security in a smart home.

Q. Introduction:

Uninformed Search is a search technique that explores possible states without using additional knowledge about which state is closer to the goal. A robot placed in an unknown maze must explore different paths until it finds the exit.

Two important uniformed search stragies are Uniform cost search (UCS) and Iterative deepening Search (IDS). These algorithms help the robot explore the maze systematically.

1. Uniform cost search (UCS)

Uniform cost search expands the node with the lowest total path cost from the starting point. It uses a priority queue to select the path with the smallest accumulated cost.

Working:

1. The robot starts at the initial position in the maze.
2. It adds the available neighboring positions to a priority queue.
3. It selects the position with the lowest total path cost.
4. It continues exploring neighboring positions and updating their costs.

2. Iterative Deepening Search (IDS)

Iterative deepening search repeatedly performs depth-limited search, increasing the depth limit each time. It combines the low memory usage of depth-first search with the completeness of breadth-first search with the completeness of finite.

working:

1. The robot starts searching with a depth limit of 0.
2. If the exit is not found, it increases the limit to 1.
3. It explores all paths up to the current depth limit.
4. The limits is increased repeatedly until the exit is found.
5. The robot returns the path to the exit.

Time and Space complexity

Let:

b = branching factor, or the average number of successors per position.

d = depth of the shallowest goal.

c = cost of the optimal solution.

ϵ = minimum positive step cost.

Most effective combination

For a robot exploring an unknown maze, a model based goal-oriented agent using Iterative deepening search is a good choice when every movement has equal cost and memory is limited.

The Model-based agent stores information about visited positions and detected walls. The goal-based component defines reaching the exit as the objective. IDS explores paths by gradually increasing the depth limit and uses relatively little memory.

However, if different paths have different movement cost, a mode-based goal-oriented agent using uniform cost search is more appropriate because UCS finds the least-cost path under standard non-negative cost assumption

Conclusion:

Uniform cost search and Iterative deepening search are useful uniformed search strategies for solving maze problems. UCS is suitable for finding the least-cost route, whereas IDS is suitable for finding the least-cost route, whereas IDS is suitable for finding a shallow solution with limited memory when movement costs are equal.